<u>Assignment 6</u>

Name: Md. Miju Ahmed
ID: 2010676104
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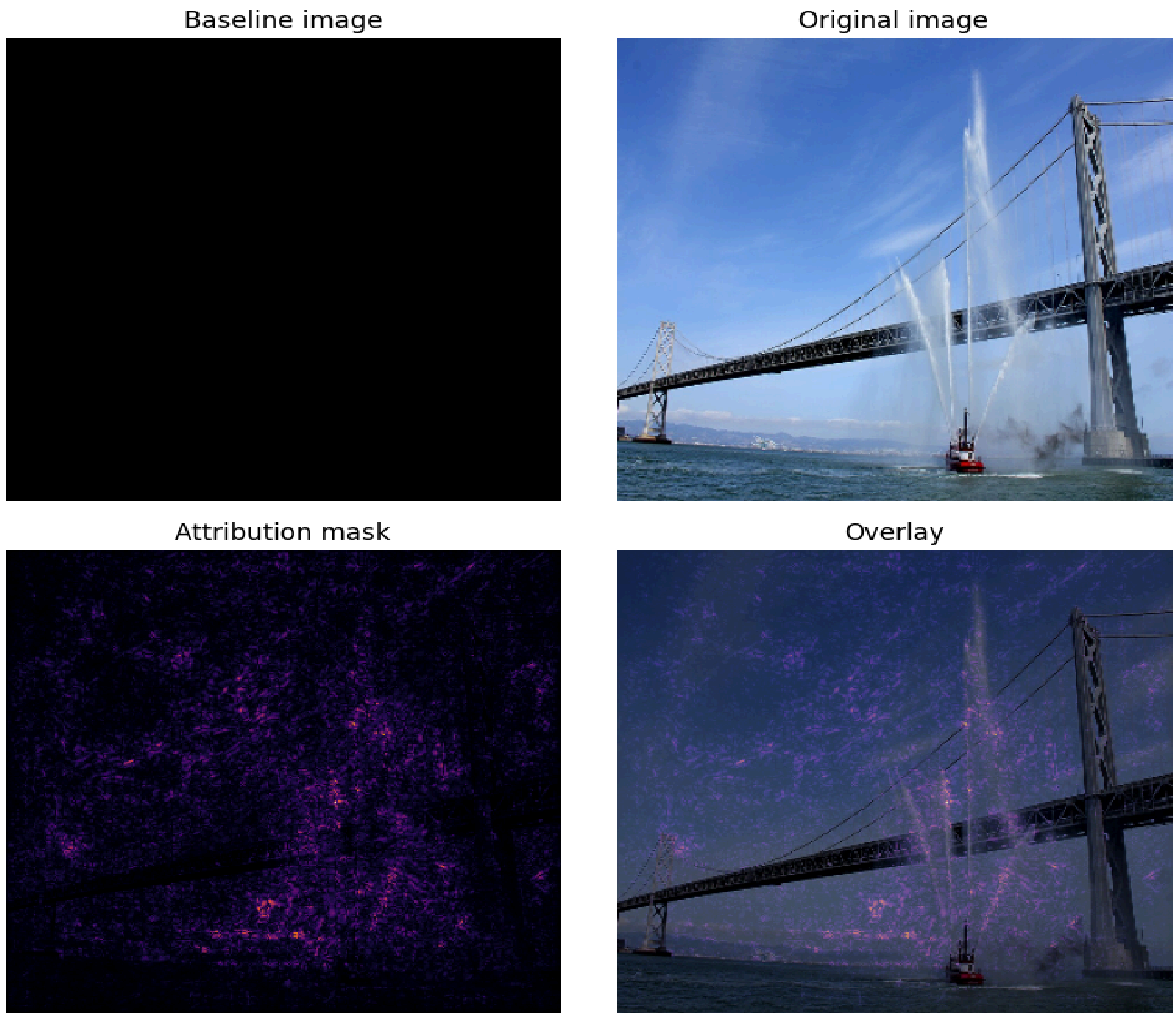
Dept: Computer Science and Engineering

Below given the investigate result for the Inception_V3 neural network for the Integrated Gradients:

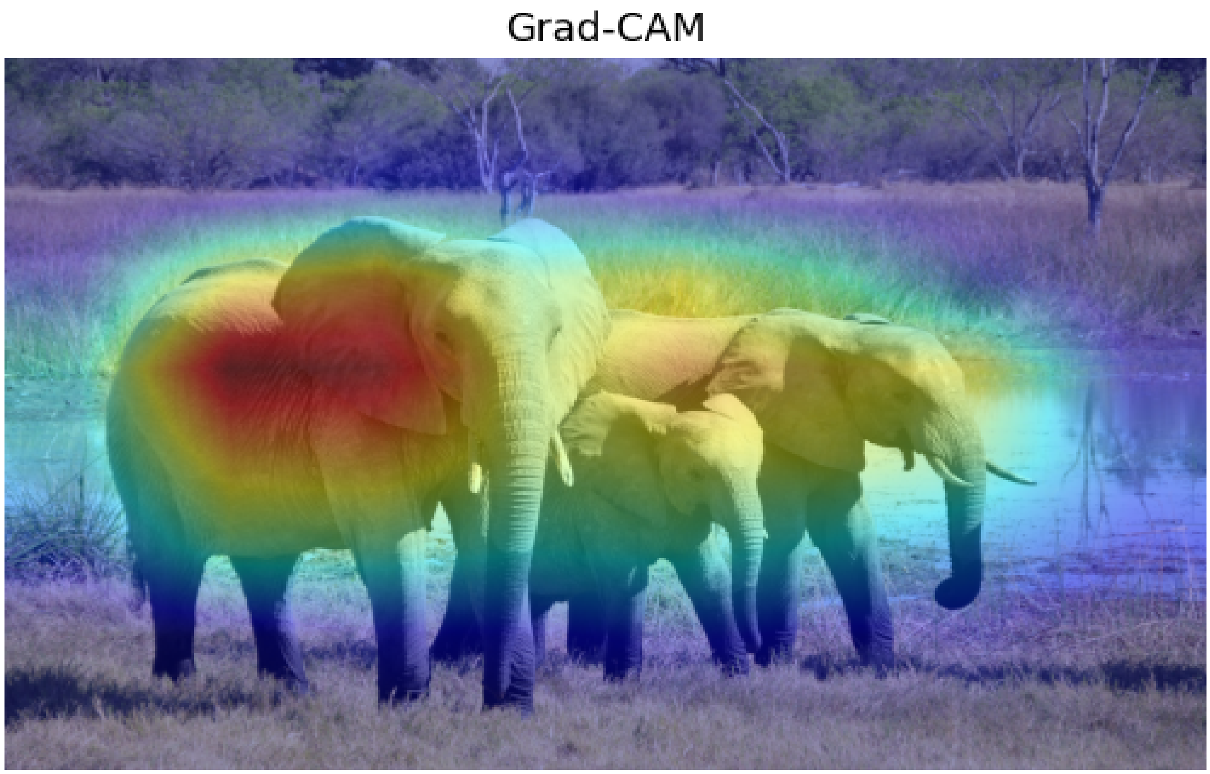
The output of the IG



After using the IG it tells us that the :
fireboat: 69.65%
pier: 9.40%
suspension_bridge: 0.79%
fountain: 0.44%
steel_arch_bridge: 0.32%
Its confidence for the fireboat is maximum and that is 69.65%. But it has not select the boat area. In this model it select the water flowing are.

Below given the investigation result for the InceptionResNetV2 neural network for Grad-CAM:

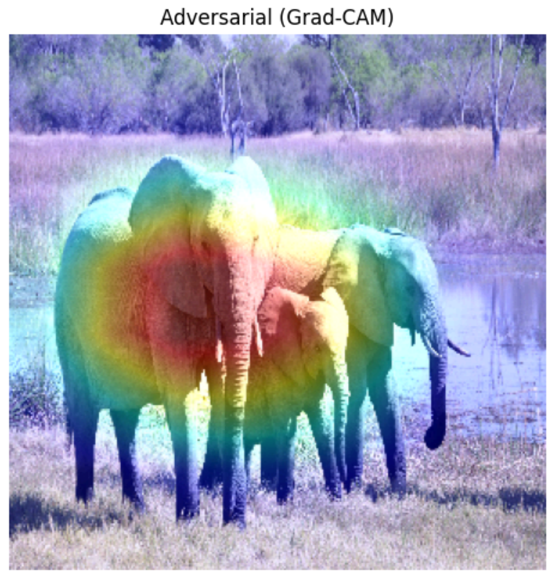
The output of the Grad-CAM



After using the Grad-CAM XAI, it tells us:
African_elephant: 78.2%
tusker: 15.2%
Indian_elephant: 0.5%
zebra: 0.0%
ladle: 0.0%

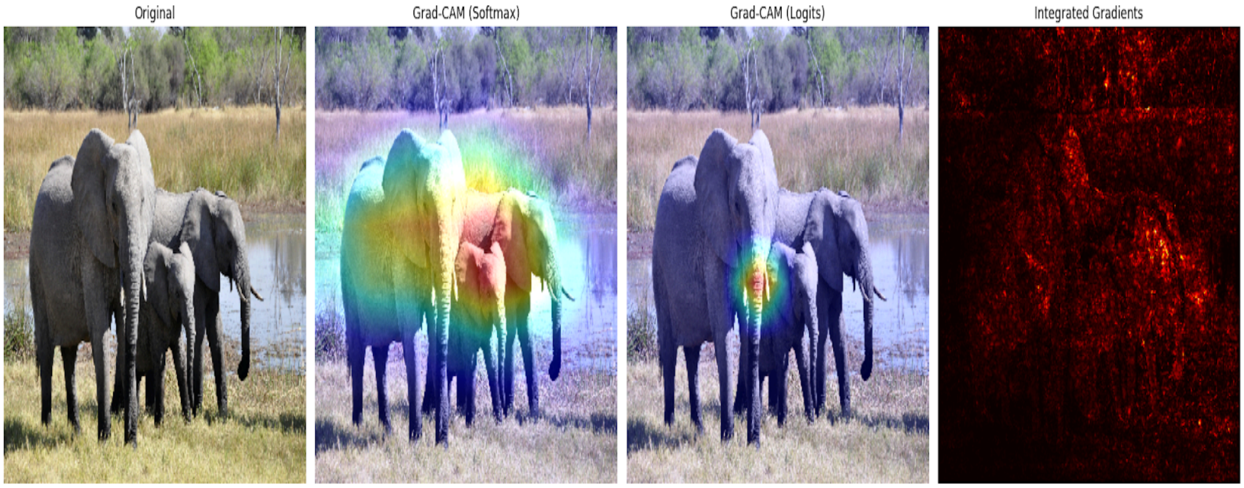
Its confidence level for the African Elephant is maximum and that is 78.2%. It selects the belly of the elephant. But we can recognize the elephant by its trunk. But it's not selected.

Below given the comparison area of the original images and the adversarial attacked images investigated area:



In the original images the Grad-CAM investigates using the ear and the head of the elephant. But in the adversarial attacked images, the Grad-CAM investigates both the ear and the trunk. Its more accurate than the investigation of the original images.

Below given the investigation area for the Grad-CAM using the softmax and the logits:



For using the softmax its results are diffuse and less localized. After using the Logits, its results are sharper and more focused than using the softmax. And the Integrated Gradient(IG) shows the pixel-wise importance of the images.

GitHub code link